Vanilla Interest Rate Swaps, Daily FRAs, and the DV01 Ladder

A desk primer on curve construction, fixing risk, and why the front stub dominates short-dated books

Prepared as a working reference · notation follows standard single-curve / multi-curve conventions

1 The vanilla swap and its floating leg

A vanilla fixed-for-floating interest rate swap (IRS) exchanges a stream of fixed coupons for a stream of floating coupons indexed to a reference rate (EURIBOR, SONIA, SOFR, ...). The economic content of the swap lives almost entirely in the *floating leg*, because that is where rate uncertainty enters.

1.1 Present value of each leg

Let $P(0,t)$ be the discount factor from today to time t (in the multi-curve world this comes from the OIS/discount curve). For a fixed leg paying rate K on accrual periods $[T_{i-1}, T_i]$ with year fractions τ_i:

$$\mathrm{PV}_{\mathrm{fix}} \;=\; K \sum_{i=1}^{n} \tau_i \, P(0,T_i).$$

The floating leg pays, on each period, the reference forward rate F_i fixed at the start of that period:

$$\mathrm{PV}_{\mathrm{flt}} \;=\; \sum_{i=1}^{m} \tau_i \, F_i \, P(0,T_i),$$

where F_i is the (simply-compounded) forward rate for $[T_{i-1}, T_i]$ implied by the *projection* (forwarding) curve. This is the key structural fact:

> *The floating leg is a strip of forward-rate agreements (FRAs), one per fixing period. Each floating coupon is economically a single FRA settling at T_i on the rate that fixes at T_{i-1}.*

1.2 The forward rate as a ratio of discount factors

The single-curve no-arbitrage forward rate for $[T_{i-1}, T_i]$ is

$$F_i \;=\; \frac{1}{\tau_i}\left(\frac{P(0,T_{i-1})}{P(0,T_i)} - 1\right).$$

Substituting into the floating-leg PV and telescoping recovers the classic single-curve identity

$$\mathrm{PV}_{\mathrm{flt}} \;=\; \sum_{i=1}^{m} P(0,T_i)\tau_i F_i \;=\; P(0,T_0) - P(0,T_m),$$

i.e. the floating leg is worth "receive notional at the start, pay notional at the end." In the multi-curve setting F_i is taken from the projection curve while the $P(0,\cdot)$ discount factors come from the OIS curve, and this neat telescoping no longer holds exactly — which is precisely why forwarding risk and discounting risk must be tracked separately.

1.3 RFR (OIS) swaps: genuinely daily fixings

For risk-free-rate swaps (SONIA, SOFR, STR) the floating coupon is not a single term fixing but a compound of *daily* overnight fixings within the period:

$$\text{Coupon}_i \;=\; \left[\prod_{d\in[T_{i-1},T_i]} (1 + r_d\,\delta_d) - 1\right],$$

where r_d is the overnight fix on business day d and δ_d its day-count. Here the "daily" structure is real: each business day locks in one more factor of the product.

2 Where "daily FRAs" actually come from

For a term index such as EURIBOR, the rate itself fixes per period (3M, 6M), not daily. The daily granularity that desks talk about comes from *curve construction*, not from the index:

1. Liquid instruments (FRAs, futures, swaps) are quoted at a discrete set of *pillar* / node tenors.
2. These are bootstrapped/calibrated into a continuous discount-factor curve $P(0,t)$.
3. Between pillars we *interpolate*, so a forward rate can be read off for *any* start date — including a day-by-day strip of forwards.

Those interpolated daily forwards can be risk-managed and, at liquid broken dates, traded (broken-date FRAs, IMM/serial gaps, etc.). So the "daily FRA strip" is a *representation* of the calibrated curve at daily resolution, driven by the interpolation scheme, rather than a claim that EURIBOR fixes daily.

> **Interpolation matters.** Common schemes — linear on zero rates, log-linear on discount factors, linear-forward, or monotone-convex — distribute risk between pillars differently. Two desks with identical books but different interpolation will report *different daily forwards* yet *identical pillar sums*. Always reconcile at pillar level.

3 Why fixings / resets create risk

Consider a single floating period. Before its fixing date T_{i-1}, the coupon depends on the *forward* F_i, which moves with the curve — so the period carries live delta. At the fixing, the rate locks to a realised number $\bar{F}_i$ and the coupon becomes a *known* cashflow paid at T_i.

3.1 DV01 collapse across a fix

Write the PV contribution of period i as $v_i = \tau_i F_i P(0,T_i)$. Its sensitivity to the forward is

$$\frac{\partial v_i}{\partial F_i} \;=\; \tau_i\, P(0,T_i) \;>\; 0 \qquad \text{(pre-fix)},$$

but the instant F_i fixes to the constant $\bar{F}_i$,

$$\left.\frac{\partial v_i}{\partial F_i}\right|_{\text{post-fix}} \;=\; 0.$$

The forward-rate DV01 of that bucket drops discretely to zero on the fixing date. Only *discount* risk (via $P(0,T_i)$) remains until payment. This is **fixing / reset risk**: hedge ratios shift discontinuously across the fix, and a bucket that was live yesterday is dead today.

3.2 RFR case: continuous decay instead of a jump

Because an OIS coupon compounds daily realised fixings, the unfixed portion shrinks each business day. The forward DV01 of the current period therefore *decays continuously* across the accrual window rather than collapsing in one jump — roughly in proportion to the fraction of the period still unfixed:

$$\mathrm{DV01}_i^{\mathrm{fwd}}(t) \;\propto\; \frac{T_i - t}{T_i - T_{i-1}}, \qquad t \in [T_{i-1}, T_i].$$

4 Why it dominates short-dated ($\leq$ 2Y) swaps

Reset mechanics matter everywhere, but they *dominate* the risk of short-dated books. The reason is proportional: the front fixing period is a large fraction of a short swap's total life, and a tiny fraction of a long swap's.

4.1 A back-of-envelope argument

Total floating DV01 is spread across m periods. The front period's share of forward risk is roughly

$$\text{front-stub share} \;\approx\; \frac{\tau_1 P(0,T_1)}{\sum_i \tau_i P(0,T_i)}.$$

For a 1Y semi-annual swap, $m = 2$ and the front stub is on the order of half the floating risk. For a 10Y swap, $m = 20$ and the front stub is only a few percent — the risk is dominated by far-forward rates and by discounting. Hence:

- In a 2Y-and-in book, one fixing moves your hedge materially; the front-stub interpolation choice is first-order.
- In a 10Y/30Y swap, a single reset barely registers against the total ladder; curve *shape* and discounting dominate.
- Short-dated books are also where carry/roll and the basis between the fixing and funding show up most visibly in daily P&L.

5 The full daily DV01 ladder

Reporting risk as a *full daily ladder* means one DV01 bucket per daily forward on the curve. Each trade projects its floating cashflows onto those forwards, and the bucket DV01 is the aggregate exposure of the whole book to that specific forward.

5.1 What a bucket sensitivity is

Let θ collect the calibrated pillar quantities (e.g. pillar discount factors or zero rates) and let the daily forward f_j be a function $f_j(\theta)$ fixed by the interpolation. The reported bucket DV01 to daily forward j is

$$\mathrm{DV01}_j \;=\; \frac{\partial \,\mathrm{PV}}{\partial f_j} \times 10^{-4}.$$

Because the daily forwards are interpolated off a smaller pillar set, these daily buckets are *not independent*: they are the pillar risk smeared across days by the interpolation. The economically meaningful quantity is the *sum* over a region between two pillars; the intra-region day-to-day shape is an interpolation artifact.

5.2 From daily ladder to hedges: the Jacobian

You cannot trade every daily bucket. To turn the daily ladder into "what do I trade," re-express risk against the calibrating (par) instruments. If $\mathbf{q}$ is the vector of par quotes of the hedge

instruments and $\mathbf{z}$ the vector of curve state (zero rates / forwards), the chain rule gives

$$\underbrace{\frac{\partial \mathrm{PV}}{\partial \mathbf{q}}}_{\text{par / tradeable risk}} = \underbrace{\frac{\partial \mathrm{PV}}{\partial \mathbf{z}}}_{\text{zero-node (daily) risk}} \cdot \underbrace{\left(\frac{\partial \mathbf{q}}{\partial \mathbf{z}}\right)^{-1}}_{\text{Jacobian}^{-1}}.$$

The matrix $\partial \mathbf{q}/\partial \mathbf{z}$ is the calibration Jacobian; its inverse maps granular zero-node risk into risk against tradeable pillars. Most risk engines expose both views — if you only see the daily ladder, ask for the par transform.

Practical rules for a daily ladder

- *Use it for visibility*: where reset risk sits in the front stub, how a single fix kills its bucket, and roll-down as each day shifts one node left.
- *Do not hedge it directly* — you will chase interpolation noise. Aggregate to tradeable pillars first.
- *Reconcile at pillar level* with other books/counterparties; the daily distribution is interpolation-dependent, the pillar sums are not.
- *Get the Jacobian* to convert zero-node risk into par-instrument hedges.

6 Worked intuition: 1Y vs 10Y across a fix

Take two par swaps, notional \$100m, and bump the front forward by 1bp.

	Front-stub fwd DV01	Total floating DV01	Front share
1Y swap (semi)	large	small	∼50%
10Y swap (semi)	similar in absolute terms	large	∼few %

The *absolute* front-bucket DV01 is comparable between the two swaps (it is essentially $\tau_1 P(0,T_1)\times$ notional $\times 10^{-4}$). What differs is its *share*: dividing by a much larger total for the 10Y. So when the front EURIBOR fixes and that bucket collapses, the 1Y book's risk profile changes shape dramatically while the 10Y book barely notices. This is the precise sense in which fixing risk "only really matters" at the short end.

7 Summary

- A vanilla term-index swap's floating leg is a strip of period FRAs; an RFR swap compounds genuinely daily overnight fixings.
- "Daily FRAs" for EURIBOR come from interpolating the calibrated curve to daily resolution, not from the index fixing daily.
- At a fixing, a bucket's forward DV01 collapses (discretely for term indices, continuously for RFRs), leaving only discount risk — this is reset risk.
- The effect dominates $\leq$ 2Y books because the front stub is a large share of a short swap's total sensitivity.
- A full daily DV01 ladder is for *visibility*; hedge off the pillar aggregation, reconcile at pillar level, and use the calibration Jacobian to map daily risk into tradeable instruments.

Notation note: $P(0,t)$ discount factor to t; F_i simply-compounded forward for period i; τ_i year fraction; DV01 = PV change per 1bp. Multi-curve conventions assumed throughout (projection curve for F_i, OIS curve for $P(0,\cdot)$).